Multi-species models and using species traits and phylogeny

Analysing Ecological Data with Detection Error

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From https://github.com/psolymos/lhreg/blob/master/vignettes/lhreg.Rmd

Introduction

The present document can be viewed as `vignette(topic = "lhreg", package = "lhreg")`. To install the package, use `remotes::install_github("psolymos/lhreg")`.

Based on the paper:

Sólymos, P., Matsuoka, S. M., Stralberg, D., Barker, N. K. S., and Bayne, E. M., 2018. Phylogeny and species traits predict bird detectability. Ecography, 41: 1595–1603. DOI: 10.1111/ecog.03415

Components of detectability

The manuscript refer to singing rate (SR, here we also refer to it as `phi`) and detection distance (DD, which we refer to as `tau`), which are linked to detectability as explained in this section. Solymos et al. (2013, DOI 10.1111/2041-210X.12106) describes the mathematical details of singing rate estimation based on removal sampling and effective detection distance estimation via distance sampling. These quantities define the probability of individuals of the species are available for sampling given presence (often referred to as availability, p), and the probability that an individual of that species is being detected given it produces a cue (often referred to as perceptibility, q).

$p = 1 - exp(-t\ SR)$, where t is the duration of the counting period. $q = DD^2\ (1 - exp(-r^2/DD^2))\ /\ r^2$, where r is the counting radius (i.e. not counting individuals beyond this distance from the observer). Probability of detection is the product to the two components: pq.

Variables

The `lhreg_data` is a data frame with all response variables (`logphi` is log singing rate, `logtau` is log detection distance) and trait values (unmodified and transformed versions).

```
library(lhreg)
```

```
Loading required package: Matrix

Loading required package: DEoptim

Loading required package: parallel
```

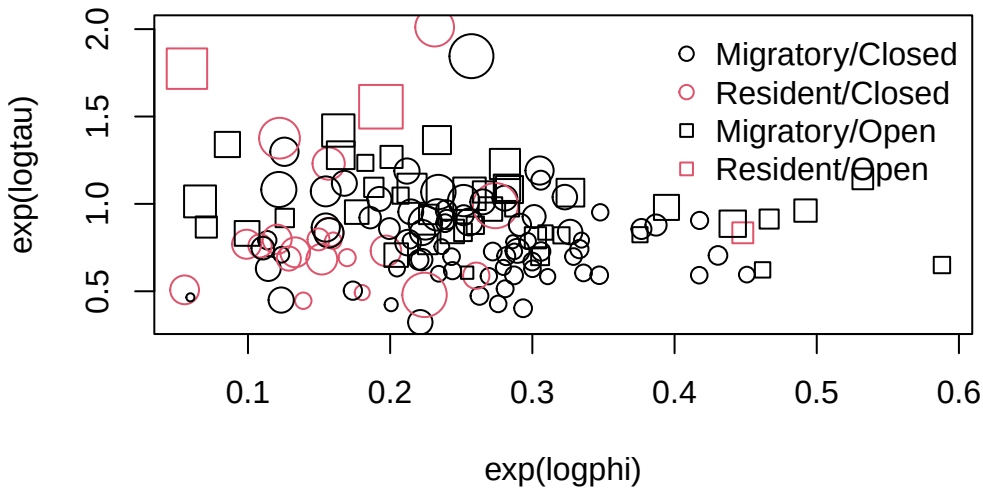
```
DEoptim package
Differential Evolution algorithm in R
Authors: D. Ardia, K. Mullen, B. Peterson and J. Ulrich

Loading required package: mvtnorm

Loading required package: pbapply
```

```r
data(lhreg_data)
str(lhreg_data)
```

```
'data.frame':   141 obs. of  24 variables:
 $ spp            : Factor w/ 141 levels "ALFL","AMCR",..: 1 2 3 4 5 6 7 8 9 10 ...
 $ scientific_name: Factor w/ 141 levels "Actitis macularius",..: 37 32 115 5 104 128 116 78
 $ common_name    : Factor w/ 141 levels "Alder Flycatcher",..: 1 2 3 4 5 6 8 7 10 9 ...
 $ WJSpecID       : int  6038 8093 12332 11885 12720 9007 12559 969 10052 13379 ...
 $ Scientific     : Factor w/ 141 levels "Actitis macularius",..: 51 32 15 5 104 122 110 89 
 $ Scientific2    : Factor w/ 141 levels "Actitis macularius",..: 51 32 15 5 104 122 110 89 
 $ logphi         : num  -1.14 -1.36 -2.18 -1.67 -1.17 ...
 $ logtau         : num  -0.1984 0.6127 -0.2329 0.0905 -0.5379 ...
 $ logphiSE       : num  0.0147 0.0157 0.0725 0.1209 0.0142 ...
 $ logtauSE       : num  0.00489 0.00553 0.00456 0.1151 0.00474 ...
 $ mass           : num  12.7 448.76 12.79 20.68 8.24 ...
 $ logmass        : num  2.54 6.11 2.55 3.03 2.11 ...
 $ MaxFreqkHz     : num  6.4 1.62 8.25 8 8 4 6.5 3.2 7 3.9 ...
 $ Migr           : Factor w/ 4 levels "Neotropical migrant",..: 1 4 4 1 1 4 4 3 1 1 ...
 $ Nesthm         : num  1 8 2 0 2 2 0 5.6 3 8 ...
 $ habitat        : Factor w/ 9 levels "Forest","Grassland",..: 7 6 6 2 1 6 6 1 3 6 ...
 $ food           : Factor w/ 8 levels "Fish","Fruit",..: 3 5 7 3 3 3 7 3 3 3 ...
 $ behavior       : Factor w/ 8 levels "Aerial Dive",..: 4 6 5 6 5 6 6 3 2 5 ...
 $ Mig            : Factor w/ 3 levels "LD","WR","SD": 1 3 3 1 1 3 3 2 1 1 ...
 $ Mig2           : Factor w/ 2 levels "M","W": 1 1 1 1 1 1 1 2 1 1 ...
 $ Hab2           : Factor w/ 2 levels "Closed","Open": 2 1 1 2 1 1 1 1 2 1 ...
 $ Hab3           : Factor w/ 3 levels "For","Open","Wood": 2 3 3 2 1 3 3 1 2 3 ...
 $ Hab4           : Factor w/ 4 levels "For","Open","Wet",..: 2 4 4 2 1 4 4 1 3 4 ...
 $ xMaxFreqkHz    : num  0.64 0.162 0.825 0.8 0.8 0.4 0.65 0.32 0.7 0.39 ...
```

```r
with(lhreg_data, plot(exp(logphi), exp(logtau),
    cex = logmass * 0.5, col = Mig2, pch = c(21, 22)[Hab2]
))
```

```r
legend("topright",
    bty = "n", pch = c(21, 21, 22, 22), col = c(1, 2, 1, 2),
    legend = c(
        "Migratory/Closed", "Resident/Closed",
        "Migratory/Open", "Resident/Open"
    )
)
```



Here is a list of species used in the manuscript:

```r
knitr::kable(lhreg_data[, c("common_name", "scientific_name", "spp")], digits = 3, row.names
```

common_name	scientific_name	spp
Alder Flycatcher	Empidonax alnorum	ALFL
American Crow	Corvus brachyrhynchos	AMCR
American Goldfinch	Spinus tristis	AMGO
American Pipit	Anthus rubescens	AMPI
American Redstart	Setophaga ruticilla	AMRE
American Robin	Turdus migratorius	AMRO
American Tree Sparrow	Spizella arborea	ATSP
American Three-toed Woodpecker	Picoides dorsalis	ATTW
Bank Swallow	Riparia riparia	BANS
Baltimore Oriole	Icterus galbula	BAOR
Barn Swallow	Hirundo rustica	BARS
Black-and-white Warbler	Mniotilta varia	BAWW
Black-billed Cuckoo	Coccyzus erythropthalmus	BBCU
Black-billed Magpie	Pica hudsonia	BBMA

common_name	scientific_name	spp
Bay-breasted Warbler	Setophaga castanea	BBWA
Black-backed Woodpecker	Picoides arcticus	BBWO
Black-capped Chickadee	Poecile atricapillus	BCCH
Belted Kingfisher	Megaceryle alcyon	BEKI
Brown-headed Cowbird	Molothrus ater	BHCO
Blue-headed Vireo	Vireo solitarius	BHVI
Blackburnian Warbler	Setophaga fusca	BLBW
Blue Jay	Cyanocitta cristata	BLJA
Blackpoll Warbler	Setophaga striata	BLPW
Boreal Chickadee	Poecile hudsonicus	BOCH
Bohemian Waxwing	Bombycilla garrulus	BOWA
Brewer's Blackbird	Euphagus cyanocephalus	BRBL
Brown Creeper	Certhia americana	BRCR
Brown Thrasher	Toxostoma rufum	BRTH
Black-throated Blue Warbler	Setophaga caerulescens	BTBW
Black-throated Green Warbler	Setophaga virens	BTNW
Blue-winged Warbler	Vermivora cyanoptera	BWWA
Canada Warbler	Cardellina canadensis	CAWA
Clay-colored Sparrow	Spizella pallida	CCSP
Cedar Waxwing	Bombycilla cedrorum	CEDW
Chipping Sparrow	Spizella passerina	CHSP
Cliff Swallow	Petrochelidon pyrrhonota	CLSW
Cape May Warbler	Setophaga tigrina	CMWA
Common Grackle	Quiscalus quiscula	COGR
Connecticut Warbler	Oporornis agilis	CONW
Common Raven	Corvus corax	CORA
Common Yellowthroat	Geothlypis trichas	COYE
Chestnut-sided Warbler	Setophaga pensylvanica	CSWA
Dark-eyed Junco	Junco hyemalis	DEJU
Downy Woodpecker	Picoides pubescens	DOWO
Dunlin	Calidris alpina	DUNL
Eastern Bluebird	Sialia sialis	EABL
Eastern Kingbird	Tyrannus tyrannus	EAKI
Eastern Phoebe	Sayornis phoebe	EAPH
Eastern Towhee	Pipilo erythrophthalmus	EATO
Eastern Wood-Pewee	Contopus virens	EAWP
European Starling	Sturnus vulgaris	EUST
Evening Grosbeak	Coccothraustes vespertinus	EVGR
Field Sparrow	Spizella pusilla	FISP
Fox Sparrow	Passerella iliaca	FOSP
Great Crested Flycatcher	Myiarchus crinitus	GCFL

common_name	scientific_name	spp
Golden-crowned Kinglet	Regulus satrapa	GCKI
Golden-crowned Sparrow	Zonotrichia atricapilla	GCSP
Gray-cheeked Thrush	Catharus minimus	GCTH
Gray Jay	Perisoreus canadensis	GRAJ
Gray Catbird	Dumetella carolinensis	GRCA
Greater Yellowlegs	Tringa melanoleuca	GRYE
Golden-winged Warbler	Vermivora chrysoptera	GWWA
Hammond's Flycatcher	Empidonax hammondii	HAFL
Hairy Woodpecker	Picoides villosus	HAWO
Hermit Thrush	Catharus guttatus	HETH
Horned Lark	Eremophila alpestris	HOLA
House Sparrow	Passer domesticus	HOSP
House Wren	Troglodytes aedon	HOWR
Indigo Bunting	Passerina cyanea	INBU
Killdeer	Charadrius vociferus	KILL
Lapland Longspur	Calcarius lapponicus	LALO
Le Conte's Sparrow	Ammodramus leconteii	LCSP
Least Flycatcher	Empidonax minimus	LEFL
Lesser Yellowlegs	Tringa flavipes	LEYE
Lincoln's Sparrow	Melospiza lincolnii	LISP
Magnolia Warbler	Setophaga magnolia	MAWA
Marsh Wren	Cistothorus palustris	MAWR
Mountain Bluebird	Sialia currucoides	MOBL
Mourning Dove	Zenaida macroura	MODO
Mourning Warbler	Geothlypis philadelphia	MOWA
Nashville Warbler	Oreothlypis ruficapilla	NAWA
Nelson's Sparrow	Ammodramus nelsoni	NESP
Northern Flicker	Colaptes auratus	NOFL
Northern Parula	Setophaga americana	NOPA
Northern Waterthrush	Parkesia noveboracensis	NOWA
Orange-crowned Warbler	Oreothlypis celata	OCWA
Olive-sided Flycatcher	Contopus cooperi	OSFL
Ovenbird	Seiurus aurocapilla	OVEN
Palm Warbler	Setophaga palmarum	PAWA
Philadelphia Vireo	Vireo philadelphicus	PHVI
Pine Grosbeak	Pinicola enucleator	PIGR
Pine Siskin	Spinus pinus	PISI
Pine Warbler	Setophaga pinus	PIWA
Pileated Woodpecker	Dryocopus pileatus	PIWO
Purple Finch	Carpodacus purpureus	PUFI
Rose-breasted Grosbeak	Pheucticus ludovicianus	RBGR

common_name	scientific_name	spp
Red-breasted Nuthatch	Sitta canadensis	RBNU
Ruby-crowned Kinglet	Regulus calendula	RCKI
Red Crossbill	Loxia curvirostra	RECR
Red-eyed Vireo	Vireo olivaceus	REVI
Rock Sandpiper	Calidris ptilocnemis	ROSA
Ruby-throated Hummingbird	Archilochus colubris	RTHU
Rusty Blackbird	Euphagus carolinus	RUBL
Ruffed Grouse	Bonasa umbellus	RUGR
Red-winged Blackbird	Agelaius phoeniceus	RWBL
Savannah Sparrow	Passerculus sandwichensis	SAVS
Scarlet Tanager	Piranga olivacea	SCTA
Sedge Wren	Cistothorus platensis	SEWR
Solitary Sandpiper	Tringa solitaria	SOSA
Song Sparrow	Melospiza melodia	SOSP
Spruce Grouse	Falcipennis canadensis	SPGR
Spotted Sandpiper	Actitis macularius	SPSA
Swamp Sparrow	Melospiza georgiana	SWSP
Swainson's Thrush	Catharus ustulatus	SWTH
Tennessee Warbler	Oreothlypis peregrina	TEWA
Townsend's Solitaire	Myadestes townsendi	TOSO
Townsend's Warbler	Setophaga townsendi	TOWA
Tree Swallow	Tachycineta bicolor	TRES
Upland Sandpiper	Bartramia longicauda	UPSA
Varied Thrush	Ixoreus naevius	VATH
Veery	Catharus fuscescens	VEER
Vesper Sparrow	Pooecetes gramineus	VESP
Warbling Vireo	Vireo gilvus	WAVI
White-breasted Nuthatch	Sitta carolinensis	WBNU
White-crowned Sparrow	Zonotrichia leucophrys	WCSP
Western Tanager	Piranga ludoviciana	WETA
Western Wood-Pewee	Contopus sordidulus	WEWP
Willow Ptarmigan	Lagopus lagopus	WIPT
Wilson's Snipe	Gallinago delicata	WISN
Wilson's Warbler	Cardellina pusilla	WIWA
Winter Wren	Troglodytes hiemalis	WIWR
Wood Thrush	Hylocichla mustelina	WOTH
White-throated Sparrow	Zonotrichia albicollis	WTSP
White-winged Crossbill	Loxia leucoptera	WWCR
Yellow-billed Cuckoo	Coccyzus americanus	YBCU
Yellow-bellied Flycatcher	Empidonax flaviventris	YBFL
Yellow-bellied Sapsucker	Sphyrapicus varius	YBSA

common_name	scientific_name	spp
Yellow Warbler	Setophaga petechia	YEWA
Yellow-headed Blackbird	Xanthocephalus xanthocephalus	YHBL
Yellow-rumped Warbler	Setophaga coronata	YRWA
Yellow-throated Vireo	Vireo flavifrons	YTVI

Here are the response variables and their standard errors:

```
lhreg_data$SR <- exp(lhreg_data$logphi)
lhreg_data$DD <- 100 * exp(lhreg_data$logtau)
knitr::kable(
    lhreg_data[, c(
        "common_name", "SR", "logphi",
        "logphiSE", "DD", "logtau", "logtauSE"
    )],
    digits = 3, row.names = FALSE
)
```

common_name	SR	logphi	logphiSE	DD	logtau	logtauSE
Alder Flycatcher	0.320	-1.138	0.015	82.006	-0.198	0.005
American Crow	0.257	-1.358	0.016	184.538	0.613	0.006
American Goldfinch	0.114	-2.175	0.073	79.222	-0.233	0.005
American Pipit	0.189	-1.669	0.121	109.467	0.090	0.115
American Redstart	0.311	-1.169	0.014	58.398	-0.538	0.005
American Robin	0.233	-1.456	0.013	93.723	-0.065	0.003
American Tree Sparrow	0.239	-1.430	0.042	96.194	-0.039	0.037
American Three-toed Woodpecker	0.157	-1.852	0.119	83.733	-0.178	0.024
Bank Swallow	0.207	-1.573	0.141	104.713	0.046	0.009
Baltimore Oriole	0.223	-1.502	0.078	83.612	-0.179	0.010
Barn Swallow	0.126	-2.072	0.176	91.939	-0.084	0.006
Black-and-white Warbler	0.244	-1.411	0.019	61.796	-0.481	0.006
Black-billed Cuckoo	0.126	-2.075	0.146	129.832	0.261	0.026
Black-billed Magpie	0.231	-1.464	0.045	201.149	0.699	0.091
Bay-breasted Warbler	0.293	-1.226	0.030	40.285	-0.909	0.009
Black-backed Woodpecker	0.197	-1.624	0.102	73.152	-0.313	0.025
Black-capped Chickadee	0.170	-1.773	0.024	69.245	-0.368	0.004
Belted Kingfisher	0.066	-2.714	0.501	101.418	0.014	0.024
Brown-headed Cowbird	0.204	-1.588	0.031	70.703	-0.347	0.007
Blue-headed Vireo	0.220	-1.514	0.029	67.926	-0.387	0.007

common_name	SR	logphi	logphiSE	DD	logtau	logtauSE
Blackburnian Warbler	0.269	-1.312	0.020	58.645	-0.534	0.008
Blue Jay	0.157	-1.853	0.021	123.055	0.207	0.005
Blackpoll Warbler	0.263	-1.336	0.065	47.355	-0.747	0.015
Boreal Chickadee	0.139	-1.974	0.059	44.585	-0.808	0.010
Bohemian Waxwing	0.055	-2.892	0.595	50.822	-0.677	0.036
Brewer's Blackbird	0.282	-1.265	0.065	109.238	0.088	0.037
Brown Creeper	0.180	-1.713	0.039	49.242	-0.708	0.009
Brown Thrasher	0.284	-1.259	0.067	108.198	0.079	0.019
Black-throated Blue Warbler	0.243	-1.414	0.049	69.802	-0.360	0.008
Black-throated Green Warbler	0.287	-1.247	0.014	72.827	-0.317	0.004
Blue-winged Warbler	0.234	-1.452	0.148	60.032	-0.510	0.056
Canada Warbler	0.300	-1.204	0.022	62.876	-0.464	0.009
Clay-colored Sparrow	0.462	-0.772	0.015	62.238	-0.474	0.011
Cedar Waxwing	0.114	-2.169	0.058	63.010	-0.462	0.006
Chipping Sparrow	0.282	-1.267	0.012	70.107	-0.355	0.003
Cliff Swallow	0.226	-1.485	0.127	76.801	-0.264	0.017
Cape May Warbler	0.276	-1.287	0.036	42.780	-0.849	0.014
Common Grackle	0.225	-1.493	0.129	89.164	-0.115	0.004
Connecticut Warbler	0.431	-0.843	0.030	70.520	-0.349	0.007
Common Raven	0.193	-1.642	0.024	155.587	0.442	0.009
Common Yellowthroat	0.309	-1.174	0.014	83.614	-0.179	0.005
Chestnut-sided Warbler	0.297	-1.215	0.011	78.740	-0.239	0.005
Dark-eyed Junco	0.222	-1.504	0.017	68.121	-0.384	0.004
Downy Woodpecker	0.109	-2.221	0.100	75.998	-0.274	0.013
Dunlin	0.086	-2.459	0.839	133.989	0.293	0.122
Eastern Bluebird	0.071	-2.649	0.522	86.761	-0.142	0.027
Eastern Kingbird	0.244	-1.412	0.081	84.235	-0.172	0.010
Eastern Phoebe	0.186	-1.682	0.135	92.117	-0.082	0.015
Eastern Towhee	0.271	-1.307	0.046	97.063	-0.030	0.024
Eastern Wood-Pewee	0.306	-1.184	0.019	113.558	0.127	0.010
European Starling	0.327	-1.118	0.068	106.409	0.062	0.003
Evening Grosbeak	0.121	-2.116	0.112	79.757	-0.226	0.016
Field Sparrow	0.183	-1.701	0.156	123.472	0.211	0.021
Fox Sparrow	0.212	-1.552	0.037	118.792	0.172	0.015
Great Crested Flycatcher	0.168	-1.785	0.051	111.781	0.111	0.009
Golden-crowned Kinglet	0.201	-1.606	0.030	42.337	-0.860	0.008
Golden-crowned Sparrow	0.201	-1.605	0.125	126.875	0.238	0.080
Gray-cheeked Thrush	0.214	-1.540	0.083	95.460	-0.046	0.025
Gray Jay	0.152	-1.884	0.026	68.387	-0.380	0.006
Gray Catbird	0.213	-1.548	0.062	77.834	-0.251	0.012
Greater Yellowlegs	0.256	-1.363	0.053	105.813	0.057	0.022

common_name	SR	logphi	logphiSE	DD	logtau	logtauSE
Golden-winged Warbler	0.214	-1.540	0.058	78.960	-0.236	0.029
Hammond's Flycatcher	0.281	-1.270	0.083	51.427	-0.665	0.021
Hairy Woodpecker	0.133	-2.016	0.063	72.297	-0.324	0.010
Hermit Thrush	0.323	-1.131	0.009	103.884	0.038	0.004
Horned Lark	0.492	-0.709	0.020	96.165	-0.039	0.014
House Sparrow	0.448	-0.803	0.054	83.446	-0.181	0.007
House Wren	0.418	-0.873	0.024	90.529	-0.099	0.011
Indigo Bunting	0.240	-1.427	0.045	91.039	-0.094	0.012
Killdeer	0.215	-1.535	0.065	108.961	0.086	0.011
Lapland Longspur	0.302	-1.197	0.109	80.618	-0.215	0.026
Le Conte's Sparrow	0.588	-0.531	0.035	65.010	-0.431	0.017
Least Flycatcher	0.417	-0.874	0.010	59.191	-0.524	0.004
Lesser Yellowlegs	0.165	-1.800	0.078	127.868	0.246	0.022
Lincoln's Sparrow	0.305	-1.186	0.018	70.078	-0.356	0.006
Magnolia Warbler	0.280	-1.274	0.015	63.650	-0.452	0.004
Marsh Wren	0.376	-0.979	0.118	82.367	-0.194	0.039
Mountain Bluebird	0.221	-1.509	0.298	32.282	-1.131	0.070
Mourning Dove	0.234	-1.453	0.049	106.783	0.066	0.006
Mourning Warbler	0.300	-1.204	0.016	66.992	-0.401	0.005
Nashville Warbler	0.335	-1.095	0.008	79.270	-0.232	0.004
Nelson's Sparrow	0.260	-1.348	0.163	87.756	-0.131	0.100
Northern Flicker	0.122	-2.106	0.052	108.218	0.079	0.008
Northern Parula	0.236	-1.443	0.029	75.542	-0.280	0.012
Northern Waterthrush	0.252	-1.380	0.029	93.509	-0.067	0.009
Orange-crowned Warbler	0.205	-1.586	0.035	63.042	-0.461	0.011
Olive-sided Flycatcher	0.281	-1.271	0.042	103.299	0.032	0.015
Ovenbird	0.387	-0.949	0.005	87.886	-0.129	0.002
Palm Warbler	0.336	-1.091	0.021	60.507	-0.502	0.009
Philadelphia Vireo	0.347	-1.058	0.052	59.176	-0.525	0.010
Pine Grosbeak	0.099	-2.312	0.305	76.915	-0.262	0.034
Pine Siskin	0.174	-1.750	0.035	50.350	-0.686	0.008
Pine Warbler	0.239	-1.433	0.032	88.971	-0.117	0.012
Pileated Woodpecker	0.122	-2.102	0.063	137.597	0.319	0.014
Purple Finch	0.110	-2.203	0.101	74.682	-0.292	0.013
Rose-breasted Grosbeak	0.256	-1.364	0.016	89.651	-0.109	0.005
Red-breasted Nuthatch	0.160	-1.833	0.026	78.936	-0.237	0.005
Ruby-crowned Kinglet	0.286	-1.251	0.013	78.320	-0.244	0.004
Red Crossbill	0.261	-1.345	0.104	58.916	-0.529	0.032
Red-eyed Vireo	0.377	-0.976	0.006	85.637	-0.155	0.002
Rock Sandpiper	0.233	-1.457	0.172	136.519	0.311	0.068
Ruby-throated Hummingbird	0.060	-2.822	0.363	46.531	-0.765	0.032

common_name	SR	logphi	logphiSE	DD	logtau	logtauSE
Rusty Blackbird	0.157	-1.852	0.117	83.421	-0.181	0.027
Ruffed Grouse	0.274	-1.295	0.031	99.020	-0.010	0.010
Red-winged Blackbird	0.394	-0.931	0.016	97.989	-0.020	0.003
Savannah Sparrow	0.467	-0.762	0.013	91.330	-0.091	0.006
Scarlet Tanager	0.192	-1.648	0.034	102.820	0.028	0.013
Sedge Wren	0.286	-1.253	0.085	96.805	-0.032	0.043
Solitary Sandpiper	0.099	-2.309	0.169	83.096	-0.185	0.029
Song Sparrow	0.291	-1.234	0.021	88.024	-0.128	0.004
Spruce Grouse	0.224	-1.495	0.235	47.933	-0.735	0.055
Spotted Sandpiper	0.176	-1.734	0.101	95.297	-0.048	0.029
Swamp Sparrow	0.251	-1.382	0.029	83.775	-0.177	0.008
Swainson's Thrush	0.327	-1.119	0.009	83.893	-0.176	0.003
Tennessee Warbler	0.451	-0.797	0.008	59.540	-0.519	0.003
Townsend's Solitaire	0.123	-2.093	0.353	44.970	-0.799	0.027
Townsend's Warbler	0.123	-2.092	0.144	70.887	-0.344	0.022
Tree Swallow	0.227	-1.482	0.055	93.982	-0.062	0.007
Upland Sandpiper	0.164	-1.811	0.120	141.986	0.351	0.039
Varied Thrush	0.252	-1.380	0.034	101.896	0.019	0.011
Veery	0.265	-1.327	0.013	101.160	0.012	0.005
Vesper Sparrow	0.532	-0.630	0.017	114.315	0.134	0.020
Warbling Vireo	0.333	-1.099	0.028	74.126	-0.299	0.008
White-breasted Nuthatch	0.150	-1.900	0.058	79.684	-0.227	0.014
White-crowned Sparrow	0.266	-1.326	0.023	106.838	0.066	0.013
Western Tanager	0.290	-1.240	0.026	73.146	-0.313	0.007
Western Wood-Pewee	0.306	-1.184	0.049	72.781	-0.318	0.016
Willow Ptarmigan	0.057	-2.863	0.514	177.379	0.573	0.158
Wilson's Snipe	0.281	-1.270	0.021	122.850	0.206	0.010
Wilson's Warbler	0.254	-1.370	0.032	60.642	-0.500	0.012
Winter Wren	0.348	-1.056	0.014	95.182	-0.049	0.004
Wood Thrush	0.305	-1.187	0.038	119.083	0.175	0.013
White-throated Sparrow	0.301	-1.200	0.007	92.758	-0.075	0.002
White-winged Crossbill	0.129	-2.050	0.052	68.731	-0.375	0.008
Yellow-billed Cuckoo	0.155	-1.866	0.176	107.157	0.069	0.069
Yellow-bellied Flycatcher	0.272	-1.301	0.021	72.750	-0.318	0.007
Yellow-bellied Sapsucker	0.155	-1.866	0.024	86.465	-0.145	0.005
Yellow Warbler	0.329	-1.112	0.020	69.862	-0.359	0.005
Yellow-headed Blackbird	0.441	-0.819	0.055	88.653	-0.120	0.104
Yellow-rumped Warbler	0.287	-1.248	0.008	59.107	-0.526	0.002
Yellow-throated Vireo	0.199	-1.612	0.062	85.886	-0.152	0.042

Other trait variables used are predictors:

common_name	mass	MaxFreqkHz	Mig2	Hab2	Nesthm
Alder Flycatcher	12.70	6.40	M	Open	1.0
American Crow	448.76	1.62	M	Closed	8.0
American Goldfinch	12.79	8.25	M	Closed	2.0
American Pipit	20.68	8.00	M	Open	0.0
American Redstart	8.24	8.00	M	Closed	2.0
American Robin	78.50	4.00	M	Closed	2.0
American Tree Sparrow	17.83	6.50	M	Closed	0.0
American Three-toed Woodpecker	55.05	3.20	W	Closed	5.6
Bank Swallow	12.68	7.00	M	Open	3.0
Baltimore Oriole	32.83	3.90	M	Closed	8.0
Barn Swallow	17.91	7.00	M	Open	2.0
Black-and-white Warbler	10.90	9.00	M	Closed	0.0
Black-billed Cuckoo	50.90	2.50	M	Closed	1.0
Black-billed Magpie	217.48	6.00	W	Closed	4.0
Bay-breasted Warbler	11.80	9.00	M	Closed	4.0
Black-backed Woodpecker	69.24	12.00	W	Closed	4.0
Black-capped Chickadee	10.80	3.20	W	Closed	2.0
Belted Kingfisher	148.00	9.00	M	Open	4.0
Brown-headed Cowbird	40.26	12.00	M	Open	2.0
Blue-headed Vireo	15.30	6.00	M	Closed	3.0
Blackburnian Warbler	9.74	11.50	M	Closed	9.0
Blue Jay	88.00	3.20	W	Closed	4.0
Blackpoll Warbler	11.84	10.22	M	Closed	2.0
Boreal Chickadee	9.80	7.70	W	Closed	3.0
Bohemian Waxwing	54.41	10.00	W	Closed	8.0
Brewer's Blackbird	62.48	6.70	M	Open	0.0
Brown Creeper	8.10	8.50	W	Closed	3.0
Brown Thrasher	68.80	6.50	M	Open	1.0
Black-throated Blue Warbler	10.14	6.00	M	Closed	0.0
Black-throated Green Warbler	8.69	8.00	M	Closed	5.0
Blue-winged Warbler	8.90	6.80	M	Closed	0.0
Canada Warbler	10.04	7.39	M	Closed	0.0
Clay-colored Sparrow	11.20	5.50	M	Open	1.0
Cedar Waxwing	31.58	8.50	M	Closed	2.0
Chipping Sparrow	12.20	5.98	M	Closed	2.0
Cliff Swallow	21.60	7.00	M	Open	4.0
Cape May Warbler	10.04	8.65	M	Closed	11.0
Common Grackle	105.18	4.00	M	Closed	3.0
Connecticut Warbler	13.30	16.00	M	Closed	0.0

common_name	mass	MaxFreqkHz	Mig2	Hab2	Nesthm
Common Raven	927.97	4.00	W	Open	17.0
Common Yellowthroat	9.54	6.30	M	Open	0.0
Chestnut-sided Warbler	9.29	7.50	M	Closed	1.0
Dark-eyed Junco	19.50	8.20	M	Closed	0.0
Downy Woodpecker	25.55	4.25	W	Closed	6.0
Dunlin	51.89	3.50	M	Open	0.0
Eastern Bluebird	27.50	3.50	M	Open	3.0
Eastern Kingbird	39.85	8.20	M	Open	3.0
Eastern Phoebe	19.70	5.00	M	Closed	2.0
Eastern Towhee	40.03	9.00	M	Open	1.5
Eastern Wood-Pewee	13.90	5.00	M	Closed	7.0
European Starling	77.14	6.00	M	Open	6.0
Evening Grosbeak	57.30	4.20	W	Closed	12.0
Field Sparrow	12.50	4.10	M	Open	1.0
Fox Sparrow	33.25	7.00	M	Closed	0.0
Great Crested Flycatcher	32.10	3.50	M	Closed	3.0
Golden-crowned Kinglet	6.19	9.50	M	Closed	10.0
Golden-crowned Sparrow	31.99	5.50	M	Open	0.0
Gray-cheeked Thrush	31.58	7.50	M	Closed	2.0
Gray Jay	71.58	1.80	W	Closed	4.0
Gray Catbird	35.30	9.00	M	Closed	1.0
Greater Yellowlegs	161.74	3.45	M	Open	0.0
Golden-winged Warbler	8.74	8.50	M	Closed	0.0
Hammond's Flycatcher	10.44	7.00	M	Closed	8.0
Hairy Woodpecker	62.65	5.50	W	Closed	8.0
Hermit Thrush	30.10	6.50	M	Closed	0.0
Horned Lark	33.33	7.00	M	Open	0.0
House Sparrow	26.51	5.00	W	Open	4.0
House Wren	10.85	8.00	M	Closed	2.0
Indigo Bunting	14.69	8.00	M	Closed	1.0
Killdeer	96.44	2.58	M	Open	0.0
Lapland Longspur	27.84	5.00	M	Open	0.0
Le Conte's Sparrow	13.00	11.00	M	Open	0.0
Least Flycatcher	10.00	7.50	M	Closed	6.0
Lesser Yellowlegs	77.50	2.79	M	Open	0.0
Lincoln's Sparrow	16.60	7.00	M	Open	0.0
Magnolia Warbler	8.14	7.50	M	Closed	1.0
Marsh Wren	10.80	9.00	M	Open	1.0
Mountain Bluebird	29.60	3.00	M	Closed	8.0
Mourning Dove	118.93	0.47	M	Closed	2.0
Mourning Warbler	11.74	5.46	M	Closed	0.0

common_name	mass	MaxFreqkHz	Mig2	Hab2	Nesthm
Nashville Warbler	8.09	9.25	M	Closed	0.0
Nelson's Sparrow	15.11	7.30	M	Open	0.0
Northern Flicker	131.46	3.35	M	Closed	7.0
Northern Parula	7.84	8.00	M	Closed	9.0
Northern Waterthrush	16.30	9.00	M	Closed	0.0
Orange-crowned Warbler	9.19	8.50	M	Closed	0.0
Olive-sided Flycatcher	32.10	3.80	M	Closed	8.0
Ovenbird	18.80	8.00	M	Closed	0.0
Palm Warbler	10.30	8.00	M	Closed	0.0
Philadelphia Vireo	11.50	7.00	M	Closed	9.0
Pine Grosbeak	56.40	4.50	W	Closed	4.0
Pine Siskin	12.70	8.00	M	Closed	5.0
Pine Warbler	11.79	5.70	M	Closed	12.0
Pileated Woodpecker	286.59	6.00	W	Closed	10.0
Purple Finch	23.30	6.00	M	Closed	4.0
Rose-breasted Grosbeak	42.00	9.00	M	Closed	3.0
Red-breasted Nuthatch	9.80	8.00	W	Closed	6.0
Ruby-crowned Kinglet	6.19	7.00	M	Closed	5.0
Red Crossbill	38.29	6.00	W	Closed	8.0
Red-eyed Vireo	16.06	6.00	M	Closed	2.0
Rock Sandpiper	81.32	7.00	M	Open	0.0
Ruby-throated Hummingbird	3.09	6.00	M	Closed	4.0
Rusty Blackbird	59.57	9.50	M	Closed	1.0
Ruffed Grouse	530.91	4.00	W	Closed	0.0
Red-winged Blackbird	50.78	6.00	M	Open	1.0
Savannah Sparrow	19.97	10.00	M	Open	0.0
Scarlet Tanager	28.20	5.00	M	Closed	6.0
Sedge Wren	9.04	7.80	M	Open	0.0
Solitary Sandpiper	48.40	5.80	M	Open	7.0
Song Sparrow	21.91	9.00	M	Closed	0.0
Spruce Grouse	473.65	6.00	W	Closed	0.0
Spotted Sandpiper	40.40	5.43	M	Open	0.0
Swamp Sparrow	16.10	7.66	M	Open	2.0
Swainson's Thrush	30.30	6.05	M	Closed	2.0
Tennessee Warbler	8.90	10.00	M	Closed	0.0
Townsend's Solitaire	33.20	5.50	M	Closed	0.0
Townsend's Warbler	8.84	8.20	M	Closed	11.0
Tree Swallow	21.20	5.95	M	Open	4.0
Upland Sandpiper	158.92	4.00	M	Open	0.0
Varied Thrush	77.50	5.85	M	Closed	6.0
Veery	31.90	5.60	M	Closed	0.0

common_name	mass	MaxFreqkHz	Mig2	Hab2	Nesthm
Vesper Sparrow	25.68	6.85	M	Open	0.0
Warbling Vireo	12.67	6.90	M	Closed	8.0
White-breasted Nuthatch	21.00	2.90	W	Closed	5.0
White-crowned Sparrow	28.00	7.00	M	Open	1.0
Western Tanager	28.10	4.42	M	Closed	11.0
Western Wood-Pewee	13.10	5.25	M	Closed	8.0
Willow Ptarmigan	566.86	7.00	W	Open	0.0
Wilson's Snipe	112.94	2.40	M	Open	0.0
Wilson's Warbler	6.96	8.10	M	Open	0.0
Winter Wren	9.74	9.00	M	Closed	1.0
Wood Thrush	50.09	4.43	M	Closed	2.0
White-throated Sparrow	24.40	6.60	M	Closed	0.0
White-winged Crossbill	28.69	6.20	W	Closed	12.0
Yellow-billed Cuckoo	64.00	0.90	M	Closed	2.0
Yellow-bellied Flycatcher	11.80	4.00	M	Closed	0.0
Yellow-bellied Sapsucker	50.30	2.41	M	Closed	8.0
Yellow Warbler	10.22	6.75	M	Closed	1.0
Yellow-headed Blackbird	62.68	7.30	M	Open	0.0
Yellow-rumped Warbler	11.94	5.80	M	Closed	4.0
Yellow-throated Vireo	18.00	5.00	M	Closed	12.0

We averaged the 1000 pseudo-posterior trees from Jetz et al. (2012) with Ericson backbone to represent the phylogenetic relationships among the species. These were turned into the phylogenetic correlation matrix, `cor_matrix`.

Analysis

Linear model were used to screen trait variables, so that we compared that *full* model with the intercept only *null* model, with or without phylogenetic correlation.

Variable screening for SR

```
library(lhreg)
data(lhreg_data)
data(cor_matrix)

y <- lhreg_data$logphi
```

```r
m4 <- lm(y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab4, lhreg_data)
m3 <- lm(y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab3, lhreg_data)
m2 <- lm(y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab2, lhreg_data)
m4s <- step(m4, trace = 0)
m3s <- step(m3, trace = 0)
m2s <- step(m2, trace = 0)
AIC(m4, m3, m2, m4s, m3s, m2s)
```

```
    df      AIC
m4   9 159.2927
m3   8 159.9963
m2   7 159.5903
m4s  5 157.4045
m3s  5 157.4045
m2s  5 157.4045
```

```r
summary(m2s)
```

```
Call:
lm(formula = y ~ Mig2 + Nesthm + xMaxFreqkHz, data = lhreg_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.35699 -0.17658  0.05151  0.23901  1.06279 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.60265    0.11180  -14.34  < 2e-16 ***
Mig2W       -0.36778    0.09966   -3.69 0.000322 ***
Nesthm      -0.01553    0.01022   -1.52 0.130918    
xMaxFreqkHz  0.33360    0.14825    2.25 0.026031 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.414 on 137 degrees of freedom
Multiple R-squared:  0.1833,	Adjusted R-squared:  0.1654 
F-statistic: 10.25 on 3 and 137 DF,  p-value: 3.916e-06
```

```r
mp <- update(m2s, . ~ . - Nesthm)

amp <- anova(mp)
round(structure(100 * amp[, "Sum Sq"] / sum(amp[, "Sum Sq"]),
    names = rownames(amp)
), 1)
```

```
       Mig2 xMaxFreqkHz   Residuals
       13.4         3.6        83.0
```

```r
mp0 <- lm(y ~ 1)
summary(mp0) # null model for SR
```

```
Call:
lm(formula = y ~ 1)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.39617 -0.23821  0.06881  0.28168  0.96531 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.49625    0.03817   -39.2   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4532 on 140 degrees of freedom
```

```r
summary(mp) # full model for SR
```

```
Call:
lm(formula = y ~ Mig2 + xMaxFreqkHz, data = lhreg_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.37585 -0.18443  0.05156  0.26123  1.08980 

Coefficients:
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -1.6619     0.1053 -15.783  < 2e-16 ***
Mig2W        -0.4108     0.0960  -4.280 3.48e-05 ***
xMaxFreqkHz   0.3599     0.1479   2.432   0.0163 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.416 on 138 degrees of freedom
Multiple R-squared:  0.1696,    Adjusted R-squared:  0.1575
F-statistic: 14.09 on 2 and 138 DF,  p-value: 2.705e-06
```

```r
## evaluating interactions
mpx <- lm(y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab2 +
    logmass:xMaxFreqkHz + Hab2:xMaxFreqkHz, lhreg_data)
mpx2 <- step(mpx)
```

```
Start:  AIC=-248.78
y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab2 + logmass:xMaxFreqkHz +
    Hab2:xMaxFreqkHz

                      Df Sum of Sq    RSS     AIC
<none>                              21.561 -248.78
- xMaxFreqkHz:Hab2     1   0.32411 21.886 -248.67
- Nesthm               1   0.49472 22.056 -247.58
- Mig2                 1   0.99609 22.558 -244.41
- logmass:xMaxFreqkHz  1   1.49463 23.056 -241.33
```

```r
summary(mpx2)
```

```
Call:
lm(formula = y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab2 +
    logmass:xMaxFreqkHz + Hab2:xMaxFreqkHz, data = lhreg_data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.42164 -0.17969  0.02572  0.24807  1.03521

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)          -2.28929    0.37591  -6.090 1.14e-08 ***
```

```
logmass               0.21429    0.09721   2.204  0.02922 *
Mig2W                -0.26467    0.10677  -2.479  0.01444 *
Nesthm               -0.01806    0.01034  -1.747  0.08297 .
xMaxFreqkHz           1.73042    0.54335   3.185  0.00181 **
Hab2Open             -0.24608    0.23044  -1.068  0.28751
logmass:xMaxFreqkHz  -0.47046    0.15494  -3.036  0.00288 **
xMaxFreqkHz:Hab2Open  0.47503    0.33596   1.414  0.15972
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4026 on 133 degrees of freedom
Multiple R-squared:  0.2502,    Adjusted R-squared:  0.2107
F-statistic: 6.339 on 7 and 133 DF,  p-value: 1.962e-06
```

Variable screening for DD

```r
y <- lhreg_data$logtau
m4 <- lm(y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab4, lhreg_data)
m3 <- lm(y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab3, lhreg_data)
m2 <- lm(y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab2, lhreg_data)
m4s <- step(m4, trace = 0)
m3s <- step(m3, trace = 0)
m2s <- step(m2, trace = 0)
AIC(m4, m3, m2, m4s, m3s, m2s)
```

```
    df      AIC
m4   9 -1.18506
m3   8 -3.17346
m2   7 -5.16348
m4s  9 -1.18506
m3s  8 -3.17346
m2s  7 -5.16348
```

```r
summary(m2s)
```

```
Call:
lm(formula = y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab2,
    data = lhreg_data)
```

```
Residuals:
     Min       1Q   Median       3Q      Max 
-0.97677 -0.11551 -0.00027  0.15237  0.71312 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.557121   0.113501  -4.909 2.60e-06 ***
logmass      0.160044   0.023038   6.947 1.44e-10 ***
Mig2W       -0.142660   0.060505  -2.358  0.01982 *  
Nesthm      -0.008508   0.005919  -1.437  0.15297    
xMaxFreqkHz -0.236341   0.091005  -2.597  0.01045 *  
Hab2Open     0.133672   0.046257   2.890  0.00449 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.231 on 135 degrees of freedom
Multiple R-squared:  0.4701,	Adjusted R-squared:  0.4505 
F-statistic: 23.95 on 5 and 135 DF,  p-value: < 2.2e-16
```

```r
mt <- update(m2s, . ~ . - Nesthm)

amt <- anova(mt)
round(structure(100 * amt[, "Sum Sq"] / sum(amt[, "Sum Sq"]),
    names = rownames(amt)
), 1)
```

```
    logmass        Mig2 xMaxFreqkHz        Hab2   Residuals 
       34.2         5.8         1.8         4.4        53.8 
```

```r
mt0 <- lm(y ~ 1)
summary(mt0) # null model for DD
```

```
Call:
lm(formula = y ~ 1)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.93698 -0.17384  0.01469  0.20521  0.89255 

Coefficients:
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.19367    0.02625  -7.379 1.29e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3117 on 140 degrees of freedom
```

```r
summary(mt) # full model for DD
```

```
Call:
lm(formula = y ~ logmass + Mig2 + xMaxFreqkHz + Hab2, data = lhreg_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.01040 -0.12314 -0.00144  0.16292  0.72842 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.59682    0.11052  -5.400 2.88e-07 ***
logmass      0.16021    0.02313   6.927 1.56e-10 ***
Mig2W       -0.16261    0.05912  -2.751  0.00676 ** 
xMaxFreqkHz -0.22061    0.09070  -2.432  0.01630 *  
Hab2Open     0.15057    0.04491   3.352  0.00104 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2319 on 136 degrees of freedom
Multiple R-squared:  0.462, Adjusted R-squared:  0.4462 
F-statistic:  29.2 on 4 and 136 DF,  p-value: < 2.2e-16
```

```r
## evaluating interactions
mtx <- lm(y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab2 +
    logmass:xMaxFreqkHz + Hab2:xMaxFreqkHz, lhreg_data)
mtx2 <- step(mtx)
```

```
Start:  AIC=-403.35
y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab2 + logmass:xMaxFreqkHz + 
    Hab2:xMaxFreqkHz

                     Df Sum of Sq    RSS     AIC
```

```
- logmass:xMaxFreqkHz  1  0.000128 7.2044 -405.34
- xMaxFreqkHz:Hab2     1  0.002181 7.2064 -405.30
<none>                              7.2042 -403.35
- Nesthm               1  0.108346 7.3126 -403.24
- Mig2                 1  0.290512 7.4947 -399.77

Step:  AIC=-405.34
y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab2 + xMaxFreqkHz:Hab2

                   Df Sum of Sq    RSS     AIC
- xMaxFreqkHz:Hab2  1   0.00206 7.2064 -407.30
<none>                          7.2044 -405.34
- Nesthm            1   0.10919 7.3135 -405.22
- Mig2              1   0.29587 7.5002 -401.67
- logmass           1   2.57629 9.7806 -364.24

Step:  AIC=-407.3
y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab2

              Df Sum of Sq    RSS     AIC
<none>                     7.2064 -407.30
- Nesthm       1   0.11026 7.3167 -407.16
- Mig2         1   0.29676 7.5032 -403.61
- xMaxFreqkHz  1   0.36003 7.5664 -402.43
- Hab2         1   0.44576 7.6522 -400.84
- logmass      1   2.57607 9.7825 -366.21
```

```r
summary(mtx2)
```

```
Call:
lm(formula = y ~ logmass + Mig2 + Nesthm + xMaxFreqkHz + Hab2,
    data = lhreg_data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.97677 -0.11551 -0.00027  0.15237  0.71312

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.557121   0.113501  -4.909 2.60e-06 ***
logmass      0.160044   0.023038   6.947 1.44e-10 ***
```

```
Mig2W       -0.142660   0.060505  -2.358  0.01982 *
Nesthm      -0.008508   0.005919  -1.437  0.15297
xMaxFreqkHz -0.236341   0.091005  -2.597  0.01045 *
Hab2Open     0.133672   0.046257   2.890  0.00449 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.231 on 135 degrees of freedom
Multiple R-squared:  0.4701,    Adjusted R-squared:  0.4505
F-statistic: 23.95 on 5 and 135 DF,  p-value: < 2.2e-16
```

Mixed model selection

The following models were compared, the prefix `t` and `p` indicates `tau` (detection distance) and `phi` (singing rate):

1. `M00` as null model M_{00},
2. `M10` as phylogeny-only model $M_{\lambda 0}$,
3. `M0b` as trait-only model $M_{0\beta}$,
4. `M1b` as combined model $M_{\lambda\beta}$.

```r
met <- "DE" # can use "SANN", "Nelder-Mead" is quickest
x <- lhreg_data
vc <- cor_matrix

## model matrix definitions
X0 <- matrix(1, nrow(x), 1) # null (for both)
colnames(X0) <- "Intercept"
Xt <- model.matrix(mt) # DD
Xp <- model.matrix(mp) # SR

## fit models for tau

tM00 <- lhreg(
    Y = x$logtau, X = X0, SE = x$logtauSE, V = vc, lambda = 0,
    hessian = TRUE, method = met
)
tM10 <- lhreg(
    Y = x$logtau, X = X0, SE = x$logtauSE, V = vc, lambda = NA,
    hessian = TRUE, method = met
)
tM0b <- lhreg(
```

```r
    Y = x$logtau, X = Xt, SE = x$logtauSE, V = vc, lambda = 0,
    hessian = TRUE, method = met
)
tMlb <- lhreg(
    Y = x$logtau, X = Xt, SE = x$logtauSE, V = vc, lambda = NA,
    hessian = TRUE, method = met
)

## fit models for phi

pM00 <- lhreg(
    Y = x$logphi, X = X0, SE = x$logphiSE, V = vc, lambda = 0,
    hessian = TRUE, method = met
)
pMl0 <- lhreg(
    Y = x$logphi, X = X0, SE = x$logphiSE, V = vc, lambda = NA,
    hessian = TRUE, method = met
)
pM0b <- lhreg(
    Y = x$logphi, X = Xp, SE = x$logphiSE, V = vc, lambda = 0,
    hessian = TRUE, method = met
)
pMlb <- lhreg(
    Y = x$logphi, X = Xp, SE = x$logphiSE, V = vc, lambda = NA,
    hessian = TRUE, method = met
)
```

Profile likelihood for lambda

Profile likelihood was calculated to understand how traits affect the strength of phylogeny through the variable `lambda`. It is possible to use `lambda` > 1 but it leads to extremely low likelihoods in our case.

```r
## set up lambda values to evaluate at
lam <- seq(0, 1, by = 0.01)

## parallel computation is faster
cl <- makeCluster(4)
## load package on workers
tmp <- clusterEvalQ(cl, library(lhreg))
```

```r
object <- tMl0
clusterExport(cl, "object")
pl_tMl0 <- pbsapply(lam, function(z, ...) profile_lambda1(object, z, ...),
    cl = cl, method = met
)

object <- tMlb
clusterExport(cl, "object")
pl_tMlb <- pbsapply(lam, function(z, ...) profile_lambda1(object, z, ...),
    cl = cl, method = met
)

object <- pMl0
clusterExport(cl, "object")
pl_pMl0 <- pbsapply(lam, function(z, ...) profile_lambda1(object, z, ...),
    cl = cl, method = met
)

object <- pMlb
clusterExport(cl, "object")
pl_pMlb <- pbsapply(lam, function(z, ...) profile_lambda1(object, z, ...),
    cl = cl, method = met
)

stopCluster(cl) # close cluster
```

Leave-one-out (LOO) analysis

Leave one out cross-validation was used to see how well we could predict the values based on data from the other species, traits and phylogeny. Mean squared error (MSE) and variance components were calculated based on LOO cross validation. We also used LOO to calculate jackknife type non-parametric confidence intervals for the estimated parameters (not presented in the manuscript).

```r
n <- nrow(x) # we will do n runs

cl <- makeCluster(4) # parallel if you wish
tmp <- clusterEvalQ(cl, library(lhreg)) # load package

loo_tM00 <- t(pbsapply(1:n, loo1, object = tM00, cl = cl))
# loo_tM00 <- t(pbsapply(1:n, loo2, object=tM00, cl=cl, method=met))
```

```r
loo_tMl0 <- t(pbsapply(1:n, loo2, object = tMl0, cl = cl, method = met))
loo_tM0b <- t(pbsapply(1:n, loo1, object = tM0b, cl = cl))
# loo_tM0b <- t(pbsapply(1:n, loo2, object=tM0b, cl=cl, method=met))
loo_tMlb <- t(pbsapply(1:n, loo2, object = tMlb, cl = cl, method = met))

loo_pM00 <- t(pbsapply(1:n, loo1, object = pM00, cl = cl))
# loo_pM00 <- t(pbsapply(1:n, loo2, object=pM00, cl=cl, method=met))
loo_pMl0 <- t(pbsapply(1:n, loo2, object = pMl0, cl = cl, method = met))
loo_pM0b <- t(pbsapply(1:n, loo1, object = pM0b, cl = cl))
# loo_pM0b <- t(pbsapply(1:n, loo2, object=pM0b, cl=cl, method=met))
loo_pMlb <- t(pbsapply(1:n, loo2, object = pMlb, cl = cl, method = met))

stopCluster(cl)
```

Parametric bootstrap

We use the `simulate` method to simulate observations from a Multivariate Normal distribution according to the input object (without the observation error) to refit the model and returns simulated values and estimates.

```r
nsim <- 999
## simulated data is nice, thus quick optimization works
metQ <- "Nelder-Mead"

cl <- makeCluster(4) # parallel if you wish
tmp <- clusterEvalQ(cl, library(lhreg)) # load package

pb_tM00 <- parametric_bootstrap(tM00, nsim = nsim, method = metQ, cl = cl)
pb_tMl0 <- parametric_bootstrap(tMl0, nsim = nsim, method = metQ, cl = cl)
pb_tM0b <- parametric_bootstrap(tM0b, nsim = nsim, method = metQ, cl = cl)
pb_tMlb <- parametric_bootstrap(tMlb, nsim = nsim, method = metQ, cl = cl)

pb_pM00 <- parametric_bootstrap(pM00, nsim = nsim, method = metQ, cl = cl)
pb_pMl0 <- parametric_bootstrap(pMl0, nsim = nsim, method = metQ, cl = cl)
pb_pM0b <- parametric_bootstrap(pM0b, nsim = nsim, method = metQ, cl = cl)
pb_pMlb <- parametric_bootstrap(pMlb, nsim = nsim, method = metQ, cl = cl)

stopCluster(cl)
```

Prediction intervals

We then calculate the prediction interval for an observation conditional on the other species and the known tree (this one and the other species included), and returns the bootstrap distribution of the prediction that can be used to calculate quantile based prediction intervals.

```
cl <- makeCluster(4)
pit <- pred_int(tM0b, pb_tM0b, cl = cl)
pip <- pred_int(pMlb, pb_pMlb, cl = cl)
stopCluster(cl)

## save the results:

save(
    list = c(
        "cor_matrix", "lam", "lhreg_data", "met", "n",
        "vc", "x", "X0", "Xp", "Xt",
        "amp", "mp", "mp0", "mpx", "mpx2",
        "amt", "mt", "mt0", "mtx", "mtx2",
        "pM00", "pM0b", "pMl0", "pMlb",
        "tM00", "tM0b", "tMl0", "tMlb",
        "pl_pMl0", "pl_pMlb", "pl_tMl0", "pl_tMlb",
        "loo_pM00", "loo_pM0b", "loo_pMl0", "loo_pMlb",
        "loo_tM00", "loo_tM0b", "loo_tMl0", "loo_tMlb",
        "pb_pM00", "pb_pM0b", "pb_pMl0", "pb_pMlb",
        "pb_tM00", "pb_tM0b", "pb_tMl0", "pb_tMlb",
        "pit", "pip"
    ),
    file = "lhreg-results-DE2.rda"
)
```

Results

Load results and set some values:

```
library(lhreg)
# load(system.file("extdata", "lhreg-results-DE.rda", package = "lhreg"))
load(system.file("extdata", "lhreg-results-DE2.rda", package = "lhreg"))

Level <- 0.95
Crit <- -0.5 * qchisq(Level, 1)
ltmp <- seq(0, 1, by = 0.0001)
```

```r
## red-yl-blue
Col <- c(
    "#2C7BB6", "#6BAACF", "#ABD9E9", "#D4ECD3", "#FFFFBF", "#FED690",
    "#FDAE61", "#EA633E", "#D7191C"
)
Col1 <- Col[1]
Col2 <- rgb(171 / 255, 217 / 255, 233 / 255, 0.5) # Col[3]
Col3 <- Col[9]
Col4 <- rgb(253 / 255, 174 / 255, 97 / 255, 0.5) # Col[7]
prt <- exp(loo_tMOb[, 1:2])
prp <- exp(loo_pMlb[, 1:2])
PIt <- t(apply(exp(pit), 1, quantile, c((1 - Level) / 2, 1 - (1 - Level) / 2)))
PIp <- t(apply(exp(pip), 1, quantile, c((1 - Level) / 2, 1 - (1 - Level) / 2)))

library(plotrix)
size_fun <- function(x, Min = 0.2, Max = 1) {
    x <- x - min(x)
    x <- x / max(x)
    x <- x * (Max - Min) + Min
    x
}
col_fun <- colorRampPalette(Col) # red-yl-blue
size_col_fun <- function(x, col_fun, nbins = 100, ...) {
    x <- size_fun(x, ...)
    q <- seq(min(x), max(x), by = diff(range(x)) / nbins)
    i <- as.integer(cut(x, q, include.lowest = TRUE))
    col_fun(nbins)[i]
}
```

Table with estimates and MSE

This table has estimates, confidence intervals, MSE, ΔAIC.

```r
get_CI <- function(x, level = 0.95) {
    t(apply(x$estimates, 2, quantile, c((1 - level) / 2, 1 - (1 - level) / 2)))
}
sf <- function(z, loo, pb, type = c("wald", "jack", "boot"), level = 0.95) {
    type <- match.arg(type)
    zz <- z$summary
    a <- c((1 - level) / 2, 1 - (1 - level) / 2)
    dig <- 2
```

```r
    if (type == "wald") {
        pcut <- function(p) {
            factor(
                c("***", "**", "*", "+", "ns")[as.integer(cut(p,
                    c(1, 0.1, 0.05, 0.01, 0.001, 0),
                    include.lowest = TRUE, right = FALSE
                ))],
                levels = c("***", "**", "*", "+", "ns")
            )
        }
        out <- structure(sapply(1:nrow(zz), function(i) {
            paste0(
                round(zz[i, 1], dig), " (SE +/- ", round(zz[i, 2], dig),
                pcut(zz[i, 4]), ")"
            )
        }), names = rownames(zz))
    }
    if (type == "jack") {
        CI <- t(apply(
            rbind(zz[, 1], loo[, 3:ncol(loo), drop = FALSE]), 2,
            quantile, a
        ))
        out <- structure(sapply(1:nrow(zz), function(i) {
            paste0(
                round(zz[i, 1], dig), " (", round(CI[i, 1], dig), ", ",
                round(CI[i, 2], dig), ")"
            )
        }), names = rownames(zz))
    }
    if (type == "boot") {
        CI <- get_CI(pb, level)
        out <- structure(sapply(1:nrow(zz), function(i) {
            paste0(
                round(zz[i, 1], dig), " (", round(CI[i, 1], dig), ", ",
                round(CI[i, 2], dig), ")"
            )
        }), names = rownames(zz))
    }
    out
}
## AIC tables
aict <- AIC(tM00, tM10, tM0b, tM1b)
```

```r
aict$dAIC <- aict$AIC - min(aict$AIC)

aicp <- AIC(pM00, pMl0, pM0b, pMlb)
aicp$dAIC <- aicp$AIC - min(aicp$AIC)

cbind(aict, t(sapply(
    list(tM00, tMl0, tM0b, tMlb),
    function(z) z$summary[c("sigma", "lambda"), 1]
)))
```

```
     df       AIC     dAIC     sigma       lambda
tM00  2 72.170443 80.68514 0.3044325 0.000000e+00
tMl0  3 45.629510 54.14420 0.4122631 7.590267e-01
tM0b  6 -8.514694  0.00000 0.2203313 0.000000e+00
tMlb  7 -6.511444  2.00325 0.2203338 4.539993e-05
```

```r
cbind(aicp, t(sapply(
    list(pM00, pMl0, pM0b, pMlb),
    function(z) z$summary[c("sigma", "lambda"), 1]
)))
```

```
     df      AIC      dAIC     sigma    lambda
pM00  2 151.5213 41.061422 0.3726461 0.0000000
pMl0  3 112.1999  1.740013 0.5319832 0.8352825
pM0b  4 123.4773 13.017366 0.3275378 0.0000000
pMlb  5 110.4599  0.000000 0.4885140 0.7887666
```

```r
## MSE
SSEt <- c(
    tM00 = sum((loo_tM00[, "pred"] - tM00$Y)^2),
    tMl0 = sum((loo_tMl0[, "pred"] - tMl0$Y)^2),
    tM0b = sum((loo_tM0b[, "pred"] - tM0b$Y)^2),
    tMlb = sum((loo_tMlb[, "pred"] - tMlb$Y)^2)
)
SSEp <- c(
    pM00 = sum((loo_pM00[, "pred"] - pM00$Y)^2),
    pMl0 = sum((loo_pMl0[, "pred"] - pMl0$Y)^2),
    pM0b = sum((loo_pM0b[, "pred"] - pM0b$Y)^2),
    pMlb = sum((loo_pMlb[, "pred"] - pMlb$Y)^2)
)
MSEt <- SSEt / n
```

```r
MSEp <- SSEp / n

Type <- "boot" # can be wald, jack, boot
zzz <- list(
    M00 = sf(pM00, loo_pM00, pb_pM00, Type, Level),
    M10 = sf(pM10, loo_pM10, pb_pM10, Type, Level),
    M0b = sf(pM0b, loo_pM0b, pb_pM0b, Type, Level),
    M1b = sf(pM1b, loo_pM1b, pb_pM1b, Type, Level)
)
m <- matrix("", length(zzz[[4]]), 4)
rownames(m) <- names(zzz[[4]])
colnames(m) <- c("M00", "M10", "M0b", "M1b")
for (i in 1:4) {
    j <- match(names(zzz[[i]]), rownames(m))
    m[j, i] <- zzz[[i]]
}
m["lambda", c("M00", "M0b")] <- "0 (fixed)"
# m[m==""] <- "n/a"
m1 <- m

zzz <- list(
    M00 = sf(tM00, loo_tM00, pb_tM00, Type, Level),
    M10 = sf(tM10, loo_tM10, pb_tM10, Type, Level),
    M0b = sf(tM0b, loo_tM0b, pb_tM0b, Type, Level),
    M1b = sf(tM1b, loo_tM1b, pb_tM1b, Type, Level)
)
m <- matrix("", length(zzz[[4]]), 4)
rownames(m) <- names(zzz[[4]])
colnames(m) <- c("M00", "M10", "M0b", "M1b")
for (i in 1:4) {
    j <- match(names(zzz[[i]]), rownames(m))
    m[j, i] <- zzz[[i]]
}
m["lambda", c("M00", "M0b")] <- "0 (fixed)"
# m[m==""] <- "n/a"
m2 <- m

m1 <- rbind(m1,
    df = aicp$df, dAIC = round(aicp$dAIC, 3),
    XV_MSE = round(MSEp, 3)
)
m2 <- rbind(m2,
```

```r
    df = aict$df, dAIC = round(aict$dAIC, 3),
    XV_MSE = round(MSEt, 3)
)

print.default(m1, quote = FALSE) # SR results
```

```
                  M00                    M10                    M0b
beta_Intercept    -1.45 (-1.51, -1.39) -1.71 (-2.15, -1.24) -1.63 (-1.79, -
1.47)
beta_Mig2W                                                   -0.37 (-0.52, -
0.23)
beta_xMaxFreqkHz                                             0.37 (0.14, 0.61)
sigma             0.37 (0.33, 0.41)    0.53 (0.39, 0.64)    0.33 (0.29, 0.36)
lambda            0 (fixed)            0.84 (0.58, 0.93)    0 (fixed)
df                2                    3                    4
dAIC              41.061               1.74                 13.017
XV_MSE            0.207                0.154                0.178
                  M1b
beta_Intercept    -1.72 (-2.13, -1.33)
beta_Mig2W        -0.2 (-0.39, -0.01)
beta_xMaxFreqkHz  0.19 (-0.06, 0.43)
sigma             0.49 (0.34, 0.59)
lambda            0.79 (0.47, 0.9)
df                5
dAIC              0
XV_MSE            0.153
```

```r
print.default(m2, quote = FALSE) # DD results
```

```
                  M00                  M10                  M0b
beta_Intercept    -0.2 (-0.24, -0.15) -0.08 (-0.42, 0.26) -0.58 (-0.77, -0.39)
beta_logmass                                               0.16 (0.12, 0.2)
beta_Mig2W                                                 -0.17 (-0.29, -0.06)
beta_xMaxFreqkHz                                           -0.23 (-0.39, -0.06)
beta_Hab2Open                                              0.14 (0.07, 0.22)
sigma             0.3 (0.27, 0.34)     0.41 (0.31, 0.5)     0.22 (0.19, 0.24)
lambda            0 (fixed)            0.76 (0.44, 0.89)    0 (fixed)
df                2                    3                    6
dAIC              80.685               54.144               0
XV_MSE            0.098                0.077                0.057
                  M1b
```

```
beta_Intercept   -0.58 (-0.78, -0.38)
beta_logmass     0.16 (0.11, 0.2)
beta_Mig2W       -0.17 (-0.28, -0.05)
beta_xMaxFreqkHz -0.23 (-0.4, -0.06)
beta_Hab2Open    0.14 (0.06, 0.23)
sigma            0.22 (0.19, 0.24)
lambda           0 (0, 0)
df               7
dAIC             2.003
XV_MSE           0.054
```

Shared variation is calculated as:

```r
Var_fun <- function(MSE) {
    Var <- 100 * (MSE[1] - MSE[-1]) / MSE[1]
    Var <- c(Var, Var[1] + Var[2] - Var[3])
    names(Var) <- c("phylo", "traits", "both", "shared")
    Var
}
## SR
round(Var_fun(MSEp), 1)
```

```
 phylo traits   both shared
  25.8   14.2   25.8   14.2
```

```r
## DD
round(Var_fun(MSEt), 1)
```

```
 phylo traits   both shared
  21.5   42.2   44.9   18.8
```

Tree with SR, DD, and trait values

Load the 1000 posterior trees, pick one and plot trait values alongside the tree.

```r
library(ape)
load(system.file("extdata", "mph.rda", package = "lhreg"))
tre <- mph[[1000]] # pick one tree

ii <- match(tre$tip.label, rownames(lhreg_data))
```

```r
d2 <- lhreg_data[ii, ]
NAMES <- paste0(d2$common_name, " (", d2$spp, ")")
tre$tip.label <- NAMES
xy <- data.frame(
    x = node.depth.edgelength(tre)[1:length(tre$tip.label)],
    y = node.height(tre)[1:length(tre$tip.label)]
)
phy_pts_size <- function(z, vari, ...) {
    points(xy[, 1] + z, xy[, 2], pch = 19, cex = size_fun(vari, Min = 0.3, Max = 1.2), ...)
}
phy_pts_col <- function(z, vari, col = 1, ...) {
    points(xy[, 1] + z, xy[, 2],
        pch = 19,
        cex = size_fun(vari, Min = 0.3, Max = 1.2),
        col = size_col_fun(vari, colorRampPalette(c("#000000", col)), Min = 0.3), ...
    )
}
phy_pts_2 <- function(z, vari, cex = 0.6, col = "#000000", ...) {
    points(xy[, 1] + z, xy[, 2],
        pch = 19,
        col = c(col, "#FFFFFF")[as.integer(vari)], cex = cex, ...
    )
    points(xy[, 1] + z, xy[, 2], pch = 21, col = col, cex = cex)
}

# pdf("Fig1.pdf", width=10, height=15)
op <- par(mar = c(1, 1, 1, 1))

plot(tre, cex = 0.6, label.offset = 22, font = 1)
segments(xy[, 1], xy[, 2], xy[, 1] + 21, xy[, 2], lwd = 1, col = 1)
phy_pts_size(2, exp(d2$logphi), col = Col1)
phy_pts_size(5, exp(d2$logtau), col = Col3)
phy_pts_2(10, d2$Mig2, col = 1)
phy_pts_size(13, d2$MaxFreqkHz, col = "#4daf4a")
phy_pts_size(16, sqrt(exp(d2$logmass)), col = "#984ea3")
phy_pts_2(19, d2$Hab2, col = "#ff7f00")
text(xy[1, 1] + c(2, 5, 10, 13, 16, 19), rep(142, 6),
    c("SR", "DD", "Migr", "Pitch", "Mass", "Habitat"),
    pos = 4, srt = 90, cex = 0.65, offset = 0
)
par(op)
# dev.off()
```

Profile likelihood profiles for lambda

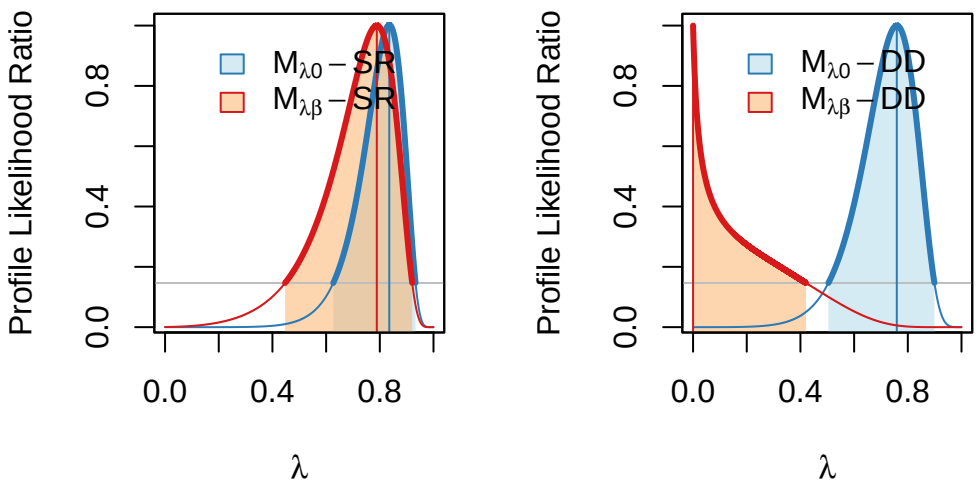
```
# pdf("FigPL.pdf", width=14, height=6)
op <- par(mfrow = c(1, 2))

Res1 <- pl_pMl0
Res2 <- pl_pMlb
yv <- exp(Res1 - max(Res1))
plot(lam, yv,
    type = "n", lwd = 1, ylim = exp(c(2 * Crit, 0)),
    xlab = expression(lambda), ylab = "Profile Likelihood Ratio"
)
tmp1 <- splinefun(lam, yv)(ltmp)
i1 <- tmp1 > exp(Crit)
yv <- exp(Res2 - max(Res2))
tmp2 <- splinefun(lam, yv)(ltmp)
i2 <- tmp2 > exp(Crit)
polygon(c(ltmp[i1], rev(ltmp[i1])),
    c(tmp1[i1], rep(-1, sum(i1))),
    col = Col2, border = NA
)
polygon(c(ltmp[i2], rev(ltmp[i2])),
    c(tmp2[i2], rep(-1, sum(i2))),
    col = Col4, border = NA
)
abline(h = exp(Crit), col = "grey")
lines(ltmp, tmp1, col = Col1, lwd = 1)
lines(ltmp[i1], tmp1[i1], col = Col1, lwd = 3)
lines(rep(ltmp[which.max(tmp1)], 2), c(1, -1), col = Col1)
lines(ltmp, tmp2, col = Col3, lwd = 1)
lines(ltmp[i2], tmp2[i2], col = Col3, lwd = 3)
lines(rep(ltmp[which.max(tmp2)], 2), c(1, -1), col = Col3)
box()
legend(0.1, 1,
    bty = "n", border = c(Col1, Col3), fill = c(Col2, Col4), pt.cex = 2, pt.lwd = 2,
    legend = c(expression(M[lambda * 0] - SR), expression(M[lambda * beta] - SR))
)
round(c(Max_Ml0 = ltmp[which.max(tmp1)], CI = range(ltmp[i1])), 3)
```

```
Max_Ml0     CI1     CI2
  0.835   0.627   0.933
```

```r
round(c(Max_Mlx = ltmp[which.max(tmp2)], CI = range(ltmp[i2])), 3)
```

```
Max_Mlx     CI1     CI2
  0.789   0.447   0.921
```

```r
Res1 <- pl_tMl0
Res2 <- pl_tMlb
yv <- exp(Res1 - max(Res1))
plot(lam, yv,
    type = "n", lwd = 1, ylim = exp(c(2 * Crit, 0)),
    xlab = expression(lambda), ylab = "Profile Likelihood Ratio"
)
tmp1 <- splinefun(lam, yv)(ltmp)
i1 <- tmp1 > exp(Crit)
yv <- exp(Res2 - max(Res2))
tmp2 <- splinefun(lam, yv)(ltmp)
i2 <- tmp2 > exp(Crit)
polygon(c(ltmp[i1], rev(ltmp[i1])),
    c(tmp1[i1], rep(-1, sum(i1))),
    col = Col2, border = NA
)
polygon(c(ltmp[i2], rev(ltmp[i2])),
    c(tmp2[i2], rep(-1, sum(i2))),
    col = Col4, border = NA
)
abline(h = exp(Crit), col = "grey")
lines(ltmp, tmp1, col = Col1, lwd = 1)
lines(ltmp[i1], tmp1[i1], col = Col1, lwd = 3)
lines(rep(ltmp[which.max(tmp1)], 2), c(1, -1), col = Col1)
lines(ltmp, tmp2, col = Col3, lwd = 1)
lines(ltmp[i2], tmp2[i2], col = Col3, lwd = 3)
lines(rep(ltmp[which.max(tmp2)], 2), c(1, -1), col = Col3)
box()
legend(0.1, 1,
    bty = "n", border = c(Col1, Col3), fill = c(Col2, Col4), pt.cex = 2, pt.lwd = 2,
    legend = c(expression(M[lambda * 0] - DD), expression(M[lambda * beta] - DD))
)
```




```
round(c(Max_Ml0 = ltmp[which.max(tmp1)], CI = range(ltmp[i1])), 3)
```

```
Max_Ml0     CI1     CI2
  0.759   0.504   0.899
```

```
round(c(Max_Mlx = ltmp[which.max(tmp2)], CI = range(ltmp[i2])), 3)
```

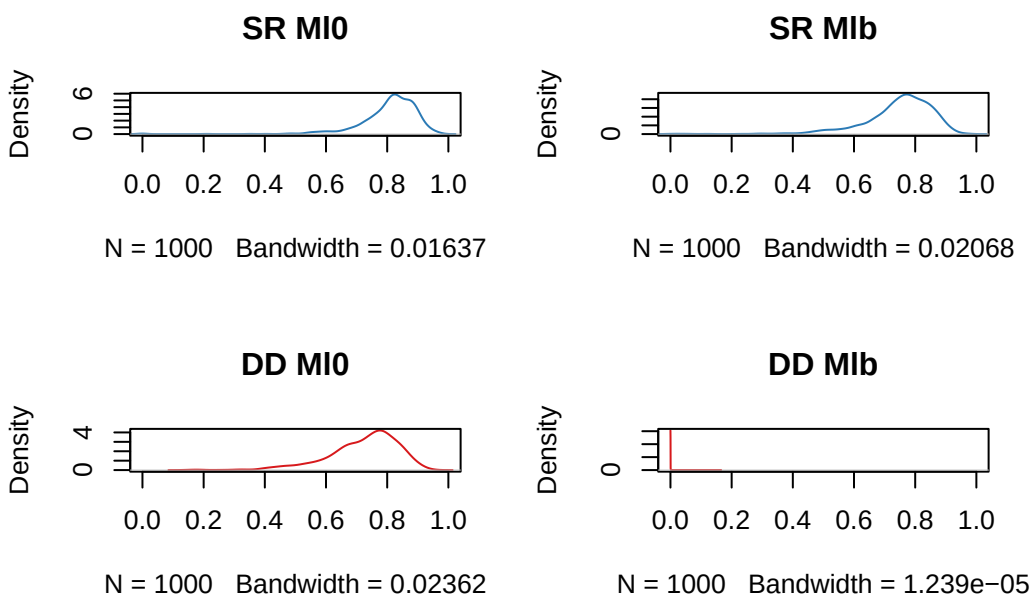
```
Max_Mlx     CI1     CI2
  0.000   0.000   0.421
```

```
par(op)
# dev.off()
```

Bootstrap distributions of lambda

```
# pdf("FigPB.pdf", width=10, height=10)
op <- par(mfrow = c(2, 2))
plot(density(pb_pMl0$estimates[, "lambda"]),
    xlim = c(0, 1), main = "SR Ml0", col = Col1
)
plot(density(pb_pMlb$estimates[, "lambda"]),
    xlim = c(0, 1), main = "SR Mlb", col = Col1
)
plot(density(pb_tMl0$estimates[, "lambda"]),
    xlim = c(0, 1), main = "DD Ml0", col = Col3
```

```
)
plot(density(pb_tMlb$estimates[, "lambda"]),
    xlim = c(0, 1), main = "DD Mlb", col = Col3
)
```



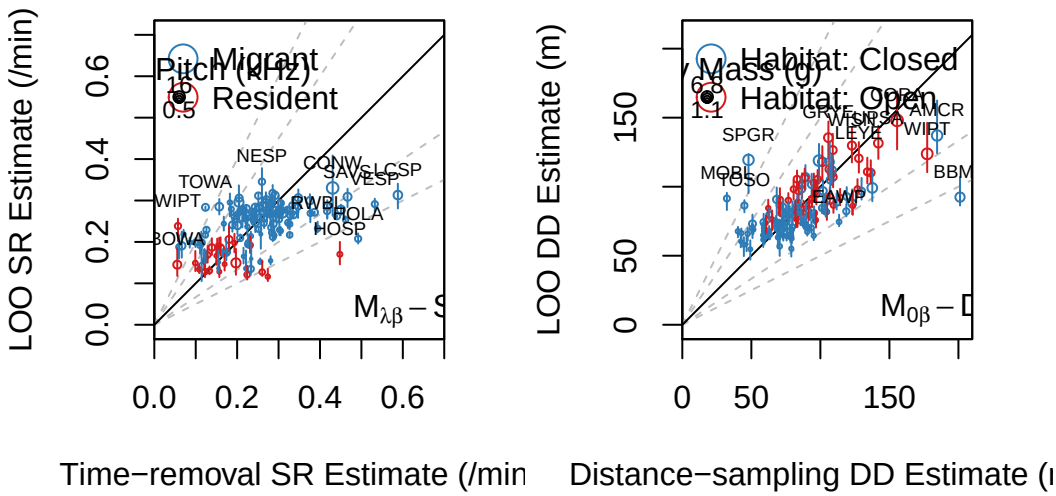

```
par(op)
# dev.off()
```

LOO cross validation plots

```
# pdf("Fig2.pdf", width=12.75, height=7)
op <- par(mfrow = c(1, 2))

Max <- 0.7
D <- as.matrix(dist(prp))
diag(D) <- Inf
D <- apply(D, 1, min)
plot(prp,
    xaxs = "i", yaxs = "i", type = "n", asp = 1,
    ylim = c(0, Max), xlim = c(0, Max),
    xlab = "Time-removal SR Estimate (/min)",
    ylab = "LOO SR Estimate (/min)"
)
```

```r
abline(0, 2, lty = 2, col = "grey")
abline(0, 1 / 2, lty = 2, col = "grey")
abline(0, 1.5, lty = 2, col = "grey")
abline(0, 1 / 1.5, lty = 2, col = "grey")
abline(0, 1, lty = 1, col = 1)
r0 <- size_fun(x$MaxFreqkHz, 0.2 * Max / 50, Max / 50)
draw.ellipse(prp[, 1], prp[, 2], r0, r0,
    border = c(Col1, Col3)[as.integer(x$Mig2)]
)
segments(prp[, 1], PIp[, 1], prp[, 1], PIp[, 2],
    col = c(Col1, Col3)[as.integer(x$Mig2)], lwd = 1
)
box()
legend("topleft",
    pch = 21, col = c(Col1, Col3),
    pt.cex = c(2, 2), bty = "n",
    legend = c("Migrant", "Resident")
)
r <- seq(0.2 * Max / 50, Max / 50, len = 5)
draw.ellipse(rep(0.06, 5), 0.55 + r - max(r), r, r, border = 1)
text(c(0.09, 0.06, 0.06), c(0.61, 0.52, 0.58),
    c("Song Pitch (kHz)", round(range(x$MaxFreqkHz), 1)),
    cex = c(1, 0.75, 0.75)
)
text(0.9 * Max, 0.05 * Max, expression(M[lambda * beta] - SR))
# Ti <- D > 0.03
# Ti <- D > 0.03 | (prp[,1]/prp[,2] > 2 | prp[,1]/prp[,2] < 1/2)
Ti <- D > 0.03 | (prp[, 1] / prp[, 2] > 3.2 | prp[, 1] / prp[, 2] < 1 / 3.2)
text(prp[, 1], prp[, 2], ifelse(Ti, as.character(x$spp), NA),
    cex = 0.6, pos = 3, col = 1
)

Max <- 100 * 2.1
D <- as.matrix(dist(prt))
diag(D) <- Inf
D <- apply(D, 1, min)
plot(100 * prt,
    xaxs = "i", yaxs = "i", type = "n", asp = 1,
    ylim = c(0, Max), xlim = c(0, Max),
    xlab = "Distance-sampling DD Estimate (m)",
    ylab = "LOO DD Estimate (m)"
)
```

```r
abline(0, 2, lty = 2, col = "grey")
abline(0, 1 / 2, lty = 2, col = "grey")
abline(0, 1.5, lty = 2, col = "grey")
abline(0, 1 / 1.5, lty = 2, col = "grey")
abline(0, 1, lty = 1, col = 1)
r0 <- size_fun(x$logmass, 0.2 * Max / 50, Max / 50)
draw.ellipse(100 * prt[, 1], 100 * prt[, 2], r0, r0,
    border = c(Col1, Col3)[as.integer(x$Hab2)]
)
segments(100 * prt[, 1], 100 * PIt[, 1], 100 * prt[, 1], 100 * PIt[, 2],
    col = c(Col1, Col3)[as.integer(x$Hab2)], lwd = 1
)
box()
legend("topleft",
    pch = 21, col = c(Col1, Col3), pt.cex = c(2, 2), bty = "n",
    legend = c("Habitat: Closed", "Habitat: Open")
)
r <- seq(0.2 * Max / 50, Max / 50, len = 5)
S <- Max / 0.7
draw.ellipse(S * rep(0.06, 5), S * 0.55 + r - max(r), r, r, border = 1)
text(S * c(0.09, 0.06, 0.06), S * c(0.61, 0.52, 0.58),
    c("Body Mass (g)", round(range(x$logmass), 1)),
    cex = c(1, 0.75, 0.75)
)
text(0.9 * Max, 0.05 * Max, expression(M[0 * beta] - DD))
Ti <- D > 0.09 | (prt[, 1] / prt[, 2] > 2 | prt[, 1] / prt[, 2] < 1 / 2)
text(100 * prt[, 1], 100 * prt[, 2], ifelse(Ti, as.character(x$spp), NA),
    cex = 0.6, pos = 3, col = 1
)
```




```
par(op)
# dev.off()
```

The percent of species where prediction intervals overlapped the estimated values:

```
library(intrval)
## SR
## # species with overlapping PI
sum(prp[, "obs"] %[]% PIp)
```

```
[1] 25
```

```
## # species with >150% or <66% of original estimates
sum((prp[, "pred"] / prp[, "obs"]) %)(% c(3 / 2, 2 / 3))
```

```
[1] 34
```

```
## # species with >200% or <50% of original estimates
sum((prp[, "pred"] / prp[, "obs"]) %)(% c(2, 1 / 2))
```

```
[1] 12
```

```
## DD
## % species with overlapping PI
sum(prt[, "obs"] %[]% PIt)
```

```
[1] 45
```

```r
## # species with >150% or <66% of original estimates
sum((prt[, "pred"] / prt[, "obs"]) %)(% c(3 / 2, 2 / 3))
```

```
[1] 7
```

```r
## # species with >200% or <50% of original estimates
sum((prt[, "pred"] / prt[, "obs"]) %)(% c(2, 1 / 2))
```

```
[1] 3
```